

EDES 301

Adaptive Irrigation Controller Proposal Update

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Proposal vs Implementation -- Changes and Updates

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Category	Proposed	Implemented	Reason for change
Control logic Modified	Closed-loop soil moisture control Soil absorption modeling Predictive watering logic	Threshold + dynamic cooldown Severity-based cooldown (15/30 min) 5 plant profiles	<i>Modeling added complexity without real benefit. Threshold cooldown handles it well</i>
Sensors Modified	Generic soil moisture sensor Generic temp/humidity sensor Reservoir moisture sensor	Adafruit Stemma (I2C capacitive) GY-BME280 (I2C, 3.3V) Grove analog water sensor	<i>Specific parts chosen once hardware was sourced. All compatible with PocketBeagle</i>
New hardware Added	Not in proposal	TXS0108E level shifter 1N4007 flyback diode	<i>Required to safely drive 5V pump from 3.3V GPIO and protect the circuit</i>
Display & alerts Modified	LED strip visual feedback Buzzer alerts	16x2 LCD with live readings Countdown to next watering Alert system on LCD display	<i>LCD gives more info than LEDs. Buzzer kept as a planned future addition</i>
Enclosure Added	Not in proposal	Custom 3D printed enclosure Integrated pot + reservoir Electronics bay + drainage	<i>Added to make system fully self-contained and finished as a real product</i>
Autoboot Implemented	Autoboot on power-up	systemd service configured Auto-restarts on crash	<i>Implemented as proposed. Added crash recovery for extra reliability</i>

Adaptive Irrigation Controller — Software Block Diagram

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